Energy Economics 59 (2016) 275–289

Contents lists available at ScienceDirect

Energy Economics

journal homepage: www.elsevier.com/locate/eneeco

Introducing MOZLEAP: An integrated long-run scenario model of the emerging energy sector of Mozambique

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ARTICLE INFO

Article history:
Received 7 May 2015
Received in revised form 25 April 2016
Accepted 5 August 2016
Available online 22 August 2016

Classification codes:
C88
O55
Q33
Q41
Q43
Q47

Keywords:
Mozambique
Energy Sector
Energy Modeling
LEAP
Scenarios

ABSTRACT

Since recently Mozambique is actively developing its large reserves of coal, natural gas and hydropower. Against this background, we present the first integrated long-run scenario model of the Mozambican energy sector. Our model, which we name MOZLEAP, is calibrated on the basis of recently developed local energy statistics, demographic and urbanization trends as well as cross-country based GDP elasticities for biomass consumption, sector structure, vehicle ownership and energy intensity. We develop four scenarios to evaluate the impact of the anticipated surge in natural resources exploration on aggregate trends in energy supply and demand, the energy infrastructure and economic growth in Mozambique. Our analysis shows that until 2030, primary energy production is likely to increase at least six-fold, and probably much more. This is roughly 10 times the expected increase in energy demand; most of the increase in energy production is destined for export. As a result, Mozambique may well become one of the leading global producers of natural gas and coal. We discuss the opportunities and challenges that this resource wealth poses for the country.

1. Introduction

Since recently Mozambique is actively developing its large reserves of coal, natural gas and hydropower. Once developed, this could make Mozambique an important player in regional and global energy markets. The recent IEA Africa Energy Outlook refers to Mozambique as an emerging large energy producer (together with Tanzania), that soon will join the group of leading energy producers in Africa, including Nigeria, South Africa and Angola (IEA, 2014). The International Monetary Fund (IMF) recently predicted that Mozambique could become the world's third largest LNG exporter after Qatar and Australia (IMF, 2016). In addition, the country maintains the fourth-largest untapped recoverable coal reserves in the world, of which large-scale exploration started in 2011. In terms of planned hydroelectric capacity, Mozambique is in Africa second only to DR Congo. Against this background, we present in this paper a new integrated long-run scenario model of the Mozambican energy sector. To the best of our knowledge, our model is the first integrated energy modeling and future planning model for Mozambique in the energy studies literature.[1]

Our scenario model is based on newly developed and locally collected energy statistics for the recent past as well as information about the latest developments and future plans as regards the production and transformation of energy in Mozambique. These data are supplemented with demographic, macro-economic and urbanization trends as well as cross-country based GDP elasticities with respect to biomass consumption, sector structure, vehicle ownership and energy intensity. The analysis makes use of LEAP, the Long range Energy Alternatives Planning

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[1] In an unpublished ministerial report, one of us drafted a rudimentary first version of an energy scenario study for Mozambique, based on data for the period 2000–2005 (Mulder, 2007). Other (consultancy) studies on energy planning in Mozambique, using different frameworks, typically consider one subsector of the energy system, like for example the electricity sector (EdM, 2004, MoE/Norconsult, 2009, Norconsult, 2011). In his exploration of energy security dynamics for Mozambique, Chambal (2010) also focusses on the electricity subsector when analyzing energy supply and demand, and does not provide an integrated energy modeling approach.

http://dx.doi.org/10.1016/j.eneco.2016.08.010

System – an integrated modeling tool that can be used to track energy consumption, production and resource extraction in all sectors of an economy (Heaps, 2012). Hence, we name our model MOZLEAP. Our analysis fits in the literature of LEAP-based studies presenting energy planning scenarios at the country level. Recent examples include studies on China (Wang et al., 2011), Greece (Roinioti et al., 2012), Japan (Takase and Suzuki, 2011) and Taiwan (Huang et al., 2011; Yophy et al., 2011). In addition, and more often, LEAP has been used for sector-level analysis in a country or region, often focusing on the power sector (Bautista, 2012; Dagher and Ruble, 2011; Kale and Pohekar, 2014; McPherson and Karney, 2014), but also on renewable energy planning (Jun et al., 2010).

Our modelling period starts with historical trends since 2000 and subsequently covers the anticipated surge in natural resources exploration until 2030. We model energy demand by households, transport and extractive industries, as well as the sectors agriculture, manufacturing, services, government and other. Also we specify electricity demand from neighboring countries in the region, given their essential role in developing the Mozambican electricity market. As regards the supply side, we model electricity production on a project by project basis, as well as gas exploration, coal mining, mineral (heavy) sands mining and charcoal production. We use the model to explore the potential impact of the expected surge in natural source exploration on aggregate trends in energy supply and demand, the energy infrastructure and economic growth in Mozambique.

The structure of the paper is as follows. In Section 2 we present our methods: the modelling framework, the database that drives our scenario model and the scenarios itself. In Section 3 we present and discuss the main elements and results of our modeling exercise. Section 4 presents in more detail the future prospects of energy supply and demand. Section 5 concludes and discusses key policy implications.

2. Methods

2.1. Modeling framework

As mentioned in the introduction, our model makes use of the LEAP framework. LEAP is intended as a medium to long-term modeling tool, designed around the concept of long-range scenario analysis (cf. Suganthi and Samuel, 2012).[2] Our model includes a historical period that comprises the period 2000–2010, in which the model is run to test its ability to replicate known statistical data. Subsequently, our model generates multiple forward looking scenarios for the period 2011–2030. LEAP supports a wide range of different modeling methodologies. In essence, the LEAP accounting framework calculates (future) energy demand as the product of activity levels (such as GDP, population, physical production levels) and energy intensity per unit of activity. Our energy demand modeling is based on a combination of historical energy and activity level data that we collected and information on demographic and urbanization trends supplied by external sources, locally collected bottom-up information as regards future electricity distribution and cross-country econometric modeling of GDP elasticities with respect to biomass consumption, sector structure and vehicle ownership. Fig. 1 and Table 1 summarize, respectively, the structure of the MOZLEAP modelling framework and the MOZLEAP model itself. In the next section we describe this approach and its results in more detail.

On the supply side, we model electricity production, gas exploration, coal mining and mineral sands mining on a project by project basis. In addition, we develop and integrate into the LEAP framework a simple biomass model to calculate future paths of charcoal production and biomass consumption in Mozambique. On the demand side we adopt a mix of these methodologies to model energy demand by households, transport and extractive industries, as well as the sectors agriculture, manufacturing, services, government and other. Also we specify electricity demand from neighboring countries in the region, given their essential role in developing the Mozambican electricity market.

2.2. Data

Most of the energy statistics for Mozambique that we use in our analysis were collected and processed by the Directorate of Studies and Planning (DEP) of the then Ministry of Energy in Mozambique (MoE, 2012). The authors of this paper have been actively involved in this process. Underlying data were solicited and provided by a range of local institutions, including National Institute of Petroleum (INP), National Company of Hydrocarbons (ENH), Mozambique Petroleum Company (PETROMOC), Cahora Bassa Hydroelectric (HCB), Mozambique power utility (EdM), Mozambique Transmission Company (MOTRACO), National Energy trust-Fund (FUNAE), South African multinational gas and Oil company (SASOL), Matola Gas Company (MGC), Portuguese Petroleum and Gas Company (GALP), VidaGas, National Institute of Statistics (INE), Mozambique Petroleum Import (IMOPETRO) and the Ministry of Planning and Development (MPD). Historical data on consumption of traditional biomass have been estimated on the basis of combined information from national survey data published by INE and international data published by the IEA and FAO. In addition, we have made use of previously published data on the energy situation in Mozambique (Arthur et al., 2010, 2012; Atanassov et al., 2012; Chambal, 2010; Cuamba et al., 2013; Cuvilas et al., 2010; Batidzirai et al., 2006; Bucuane and Mulder, 2009; Brouwer and Falcão, 2004; Schut et al., 2010).

Data on existing and future production of mineral resources (coal, natural gas and mineral sands) were compiled on the basis of information gathered from the then Ministry of Mineral Resources (MIREME), KPMG International (2013), United States Geological Survey (Yager, 2012) and the US Energy Information Administration (EIA/DOE). In addition, we collected information from press releases by private companies (in Bloomberg, Reuters, Mining Weekly, Mozambique Information Agency-AIM, and other national press), as well as from personal communications with local experts. Information on future electricity trade in the region is based on information published in the Integrated Resource Plan by the South African government (SA Department of Energy, 2011) and interviews with local experts. Finally, demographic and economic data on Mozambique were obtained from INE, the Ministry of Planning and Development and the Mozambique Central Bank (BM) as well as from the World Bank, the International Monetary Fund (IMF, 2013), the United Nations Department of Economic and Social Affairs (2011 Revision), and the African Development Bank (AfDB). All locally collected data, insofar possible, have been checked against data from international sources, including British Petroleum (BP, 2012), International Energy Agency (IEA, 2013a, 2014), United Nations Populations statistics and the World Development Indicators as published by the World Bank.

2.3. Scenarios

Energy scenarios are self-consistent storylines of how an energy system might evolve over time. Since this is, to the best of our knowledge, the first integrated energy modeling and future planning study for Mozambique in the energy studies literature, we chose to develop in this paper a limited number of scenarios that are intentionally fairly simple and straightforward. Our main goal is to introduce our newly developed scenario model MOZLEAP, and to use it for highlighting major trends in the transformation of the emerging Mozambican energy sector, including the expected consequences for both domestic and international energy markets. The development of richer scenarios, including more detail and variation in terms of energy policies, structure

[2] For more information see www.energycommunity.org.

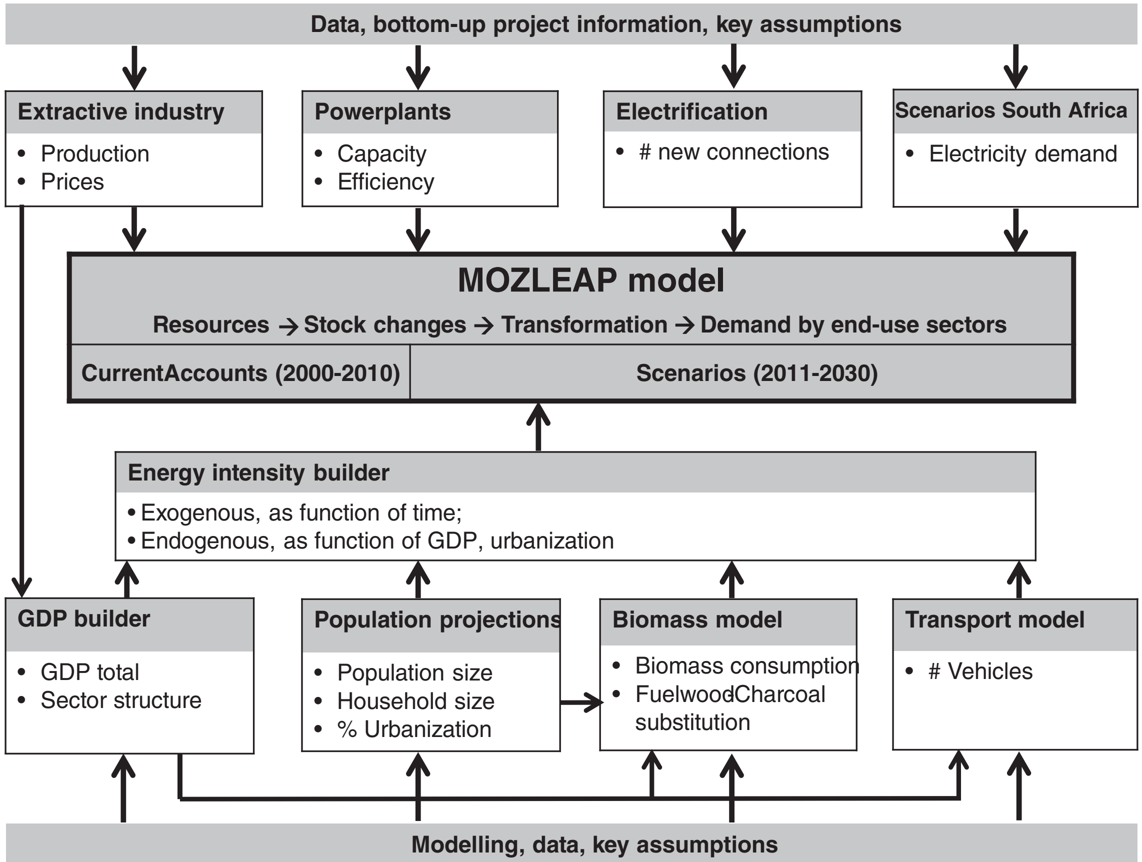

Fig. 1. Structure of the MOZLEAP modelling framework.

of energy demand, energy supply mix options and regional differences, is deliberately left for future work.

Energy outlooks usually give three basic scenarios – medium, high and low – that are often largely defined by GDP and population growth expectations. We follow this approach, but add a fourth scenario that assumes exploitation of Mozambique's natural resources exploration to its fullest potential. We label our three basic scenarios as Reference, Reference High and Reference Low. Reference is the *most likely* development path. Development of GDP in the Reference scenario is based on baseline projections plus activities of new extractive industry and electricity generation projects that are (almost) sure to be realized, taking into account realistic and somewhat conservative estimates about the output price development in the extractive and aluminum industry. Furthermore, it adopts a medium variant of population growth scenarios, a modest decline in household size, a moderate speed of urbanization and somewhat conservative estimates as regards the development of energy intensity improvements across sectors. Reference High and Reference Low then refer, respectively, to the optimistic and pessimistic variant of Reference – thus assuming higher (lower) baseline economic growth, lower (higher) population growth, higher (lower) speed of urbanization, faster (slower) decline of household size and higher (lower) output price developments in the extractive and aluminum industry. We refer to Table 2 for a brief summary and overview or scenarios.

Finally, our Extractive scenario describes the expected evolution of the Mozambican energy system if all potential projects of extractive and aluminum industries as well as power generation are realized, including those projects that are yet (very) uncertain. In other words, this scenario tells the story of the Mozambican economy and energy sector becoming very much extractive industry driven. Because of this focus, we assume all other leading dimensions of the model (population growth, household size, speed of urbanization, energy intensity improvements and output price developments) to be equal to the Reference or Reference High scenario (see Table 2). This straightforward set-up, again, is motivated by our aim to show the potential impact of an extractive industry driven development path as caused by the mere expansion of this activity rather than by (optimistic) energy intensity changes or price developments. We leave it to future work to analyze the potential impact of price volatility on international natural resource and commodity markets on the Mozambican economy and energy sector, detailing the (future) evolution of international commodity price variation across markets and sectors.

3. The scenario model

In this section we present the different parts of our scenario model (see also Fig. 1) in more detail, and show how the expected surge in natural resource exploration relates to aggregate trends in economic growth, population growth, urbanization, biomass consumption, sector structure, vehicle ownership and energy intensity in Mozambique.

3.1. GDP builder

Together with population growth, per capita GDP is a key driving force in our model. Evidently, on the one hand energy is an essential production factor that fuels economic growth, while on the other hand increasing standards of living lead to growing demand for energy (GEA, 2012). In accordance with this, our model structure assumes that across sectors growing GDP is associated with higher energy use. Also, we assume that total biomass consumption and fuel demand for road transport are determined by GDP per capita, either

Table 1
The structure of the MOZLEAP modeling framework.

Category, Sector	Subsector	Activity	Energy type
DEMAND			
Residential	Electrified	# Households	Electricity, LPG, Kerosene, Charcoal, Fuelwood
	Non Electrified	# Households	Kerosene, Charcoal, Fuelwood
Agriculture			Electricity, Diesel
Manufacturing	MOZAL	Metric Tonne	Fuel Oil, Natural Gas, Electricity
	Other Industry	GDP	Fuel Oil, Natural Gas, Electricity, Diesel
Services	Commercial Services	GDP	Electricity, LPG, Fuelwood, Charcoal
	Public Lighting	Not applicable	Electricity
Government		GDP	Electricity
Extractive Industries	Coal Mining	Metric Tonne	Electricity, Diesel
	Natural Gas Exploration	GDP	Natural Gas
	Mineral Sands Mining	Metric Tonne	Electricity, Diesel
Other Sectors		GDP	Electricity
Transport Road	Passenger Cars	# Vehicles	Gasoline, Ethanol
	Trucks	# Vehicles	Diesel, Methanol
	Motorcycles	# Vehicles	Gasoline, Ethanol
	Tractors	# Vehicles	Diesel, Methanol
Regional Electricity Demand	South Africa	Not applicable	Electricity
	Zimbabwe	Not applicable	Electricity
	Other	Not applicable	Electricity
STATISTICAL DIFFERENCES			
	Primary		All primary
	Secondary		All secondary
TRANSFORMATION			
Transmission and Distribution			Electricity, Natural Gas
Electricity Generation	Solar PV		Electricity
	Hydro		Electricity
	Thermal Natural Gas		Electricity
	Thermal Coal		Electricity
Charcoal Making	Existing		Charcoal
	New Efficient		Charcoal
Coal Mining			Coal
Natural Gas Exploration			Natural Gas
STOCK CHANGES			
	Primary		All primary
	Secondary		All secondary
RESOURCES			
	Primary		All primary
	Secondary		All secondary

directly (in the case of biomass) or indirectly (in the case of road transport, assuming that vehicle ownership is determined by per capita GDP). Finally, we assume that in various sectors of our model the evolution of energy intensity is a function of GDP growth, reflecting the notion of increasing energy efficiency under economic development (Lescaroux, 2011).

To model future development paths of GDP we developed a so-called GDP builder that is embedded in LEAP's overall accounting framework. We construct future GDP paths by combining a top-down and bottom-up approach, as follows. We start with historical data from existing sources (Mozambique Central Bank, National Statistics Institute, IMF, Worldbank) on Mozambique's total GDP and its sector structure for the historical period 2000–2010. From these data series we derive historical GDP growth rates, excluding the extractive industry. We call this baseline GDP growth. Subsequently, adopting a simple

Table 2
Scenarios for MOZLEAP.

Scenario	Variant	Description	Annual growth Baseline GDP*	New projects extractive industry and power sector	Output price extractive and aluminum industry	Population growth	Household size	Speed of urbanization	Energy intensity improvements
Reference	Medium	The *most likely* development path.	Gradual decrease to 4.7% in 2030.	Including those that are (almost) sure to be realized	Realistic and somewhat conservative estimates	Medium growth scenario	Linear extrapolation of decreasing trend	Medium scenario	Realistic and somewhat conservative estimates
	Low	The *pessimistic* variant of Reference-medium.	Gradual decrease to 3.8% in 2030.	Same as Reference-medium	Low estimates	High growth scenario	Trend 50% slower than Reference-medium	Trend 50% slower than Reference-medium	Same as Reference-medium
	High	The *optimistic* variant of Reference-medium.	Gradual decrease to 5.9% in 2030.	Same as Reference-medium	High estimates	Low growth scenario	Trend 50% faster than Reference-medium	Trend 50% faster than Reference-medium	Same as Reference-medium
Extractive		The *extractive industry driven* development path.	Same as Reference-high.	Including all planned projects, including those that are uncertain	Same as Reference-high.	Same as Reference-high.	Same as Reference-high.	Same as Reference-high.	Same as Reference-medium.

* Baseline GDP means all sectors excluding extractive industry.

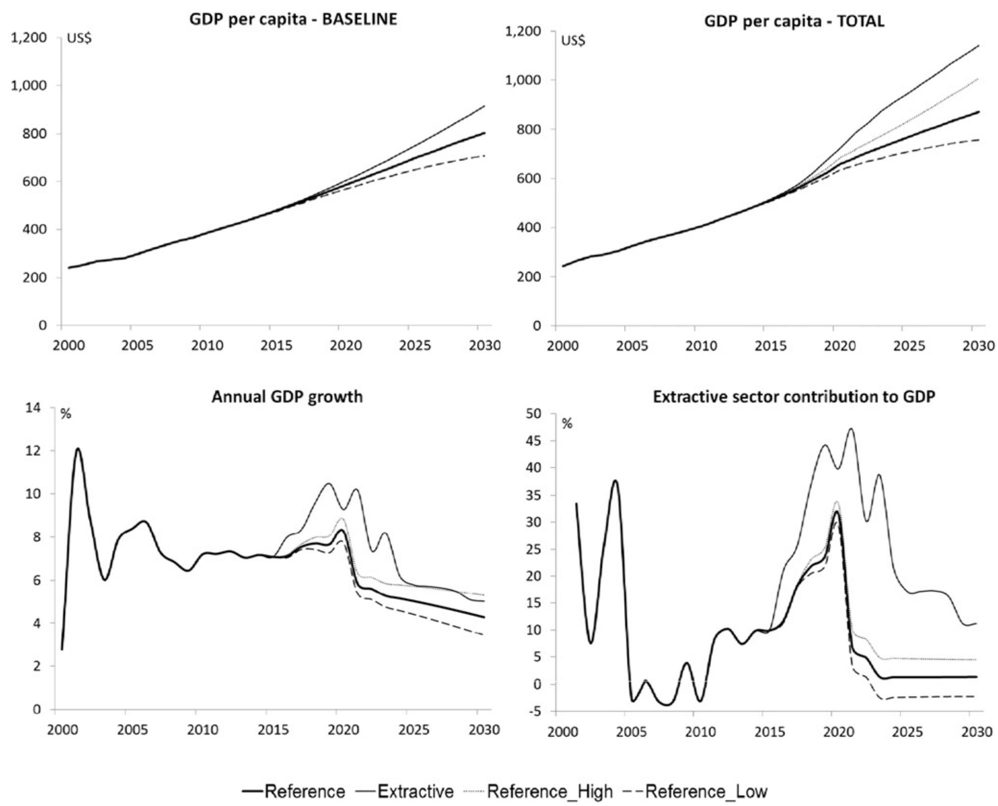

Fig. 2. Per capita GDP across scenarios; Baseline GDP (left) and Total GDP (right).

top-down approach, for the period 2011–2030 we assume that baseline GDP growth $g_t^{Y^{Base}}$ follows a logistically declining trend as function of time t, such that baseline GDP is defined as:

$$Y_t^{Base} = Y_{t-1}^{Base}\left(1 + g_{t-1}^{Y^{Base}} e^{-\delta t}\right), \tag{1}$$

with δ a parameter that determines the speed of decline in the logistic growth curve. During the period 2000–2010 Mozambique experienced rapid economic growth, on average about 7.5% per year for total GDP and 5.5% for per capita GDP. We have updated our model with the GDP growth rates for the period 2011–2015. The value of δ in Eq. (1) is scenario-specific and chosen such that annual GDP growth from 2016 gradually evolves towards 3.8% – 5.6% by 2030, depending on the scenario (see also Fig. 2 and Table 3). Note that the sharp fall in GDP growth rates and extractive sector contribution to GDP just after 2020, is mainly caused by stabilized production quantities of natural gas, after a few years of rapid expansion of natural gas production capacity (see section 4.1 and Table A.4 in the Appendix). Hence, the peak around 2020 in the two graphs at the bottom of Fig. 2 reflects a rapid transition to a higher steady state GDP level (see the top-right graph in Fig. 2), rather than a sudden decline of macroeconomic performance.

Next, using a bottom-up approach, we construct GDP separately for each extractive industry, as follows. First, based on the information in our dataset (see Section 2), we specify per existing and planned extractive industry project the expected future production in physical units, Q. We include in our model electricity production, gas exploration, coal mining, mineral sands mining and the aluminum industry, which

Table 3
Key assumptions MOZLEAP model.

			Reference												Extractive			
			Medium				Low				High							
	Unit	2010	2015	2020	2025	2030	2015	2020	2025	2030	2015	2020	2025	2030	2015	2020	2025	2030
GDP																		
Parameter δ	1/100	--	0.2	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Unit price change																		
Natural Gas	%	--	0.0	2.0	2.0	2.0	0.0	0.0	0.0	0.0	0.0	4.0	4.0	4.0	0.0	4.0	4.0	4.0
Mineral Sands	%	--	0.0	0.0	0.0	0.0	-1.0	-1.0	-1.0	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Coal	%	--	-1.0	0.0	0.0	0.0	-1.0	-2.0	-2.0	-2.0	-1.0	2.0	2.0	2.0	-1.0	2.0	2.0	2.0
Aluminum	%	--	0.0	0.0	0.0	0.0	-1.0	-1.0	-1.0	-1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Population																		
Growth population	%	2.45	2.36	2.25	2.13	2.01	2.60	2.64	2.56	2.43	2.14	1.88	1.70	1.57	2.14	1.88	1.70	1.57
Growth urban population	%	3.23	3.39	3.50	3.50	3.45	1.69	1.75	1.75	1.73	5.08	5.25	5.26	5.18	5.08	5.25	5.26	5.18
Household size	#	4.33	4.29	4.26	4.22	4.19	4.30	4.29	4.27	4.25	4.28	4.23	4.18	4.13	4.28	4.23	4.18	4.13
Electricity distribution																		
# New connections/year	1000	110	135	123	112	100	129	109	90	70	141	137	134	130	135	123	112	100
Losses*	%	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0

δ: Speed of decline logistic curve of baseline GDP growth.
* Transformation and distribution losses.

transforms imported bauxite into aluminum for export. In Section 4.1 we describe the considered extractive industry projects in more detail. Second, we calculate for each project the GDP value per physical unit of production, p. To do so, we start with historical data until 2012, which we subsequently extrapolate, assuming a simple but scenario-specific trend based on expected international market prices of the primary resources involved (LNG, mineral sands, coal, aluminum). Third, we estimate future GDP of the extractive industry by combining these price trends $g_{i,t}^{p^{Extr}}$ with expected physical production patterns per project i, and subsequently aggregating over all projects, such that:

$$Y_t^{Extr} = \sum\nolimits_i Q_{i,t}^{Extr}\left[p_{i,t-1}^{Extr}\left(1+g_{i,t}^{p^{Extr}}\right)\right]. \tag{2}$$

Together with baseline GDP this sums up to total GDP:

$$Y_t^{Total} = Y_t^{Base} + Y_t^{Extr}. \tag{3}$$

This implies that total GDP growth rate is a weighted composition of baseline and extractive GDP growth rates.

As regards the evolution of p, our extrapolation methodology assumes a simple, scenario-specific, trend based on expected international market prices of the primary resources involved (LNG, mineral sands, coal, aluminum). These prices are partly based on expert judgments for the upcoming years, published in a variety of resources (IEA, 2013b; KPMG, 2013), while for the remaining years price trends are assumed to follow a straightforward but scenario-specific pattern, with annual price fluctuations varying between −2% and 4%. Given the expected large relative size of the extractive industry in the future economy of Mozambique, future price trends for primary resources are deliberately designed to be conservative, in order to avoid an upward bias in future GDP development paths. We refer to Table 3 for further details.

Finally, we construct a sectoral breakdown of aggregate GDP by calculating future sector shares of four main sectors (agriculture, services, manufacturing and government) as percentage of total GDP. The underlying idea, of course, is that economic development typically involves a change in the sectoral composition of economies, with the industrialization process inducing a shift from the agricultural sector towards industry, followed by a deindustrialization phase increasing the importance of the service sector (e.g., Baumol, 1967; Maddison, 1991, 1999). Again, our starting point is historical data for the period 2000–2011 from existing sources. Next, we assume that the respective sector shares S evolve over time as a function of baseline GDP per capita, y_t^{Base}, according to the following logistic curve:

$$S_{(t)} = S_{t-1} * \left[1+\frac{\theta}{y_t^{Base}}\right]^{\Delta y_t^{Base}} \tag{4}$$

with parameter θ signifying the elasticity of the change in the sectoral composition of the economy under influence of economic development. The value of θ is sector-specific and is derived from cross-country regressions of the relation between per capita GDP and the respective sector share, using Worldbank data for 39 countries with per capita GDP values between US$700 and US$3000; estimated coefficients vary from −2.94 for agriculture to 4.86 for manufacturing. We refer to Table A.1 in the Appendix for details.

The results of our GDP calculations are summarized in Fig. 2. First, this figure clearly shows that Mozambique is extremely poor but at the same time experienced rapid economic growth over the last decade and a half. For the period 2000–2015 the average annual growth rate of GDP was about 7.4% (IMF, 2016).[3] In that period, per capita GDP doubled from $250 to just over $500 (the latter equals to about $1100 in PPP terms). These levels roughly correspond with 9% of the per capita GDP level in South Africa and 2% of the per capita GDP level in the USA, and imply that still about half of the Mozambican population lives below the local absolute poverty line (Boom, 2011). In addition, Fig. 2 shows that our modelling of Baseline GDP (see Eq. (1)) leads to a gradual increase of Baseline GDP per capita to levels of $700–$900 by 2030, depending on the scenario. Extractive GDP per capita is expected to increase dramatically over time, from almost zero in 2000 to $70–$90 after 2020 in the Reference scenarios, and up to $225 in the Extractive scenario. Together these developments cause total per capita GDP to be in the range of $750–1150 by 2030, depending on the scenario; this equals a 50–125% increase from 2015 levels. Third, Fig. 2 shows that GDP growth rates are expected to peak around 2020, especially in the extractive scenario. This growth acceleration is largely driven by extractive industry expansion, most notably the assumption that gas processing facilities in the Rovuma Basin, in the far north of the country, will begin production around 2020. The International Monetary Fund (IMF) recently predicted that Mozambique's average growth rate during the first half of the next decade could reach over 20% per annum (IMF, 2016). Given the political, financial and technical uncertainties in developing these gas fields, we work with more conservative assumptions. Nevertheless, we predict growth rates of over 10%, of which one third to almost half will be caused by the extractive sector expansion, depending on the scenario – as can be seen from Fig. 2. This resembles the situation in the early 2000s when the extractive sector was an important driving force of GDP growth, due to the opening up of the Mozal aluminum smelter near the capital Maputo. Together, our projections imply that over the next decade the extractive sector will comprise about 10–20% of the total economy, depending on the scenario.

3.2. Population projections

Size and growth of the population helps define critical indicators in our model, such as per capita GDP, the electrification rate, total residential energy consumption, and fuel consumption by passenger cars. In addition, these indicators are influenced by the composition of the population in terms of the urban–rural divide and whether or not households have access to electricity. Growth of population has been calculated as the product of birth, mortality and net migration statistics, based on information from the National Statistics Institute (INE) that is derived from national censuses 1997 and 2007, supplemented with data obtained from local surveys on, amongst others, infant mortality and HIV prevalence. Future projections of these various demographic statistics have been obtained by INE through a combination of extrapolating historical trends, collecting new data from local surveys (after 2007) and the use of demographic modelling software developed by the UN and the US Census Bureau. Fig. 3 and Table 3 summarize our key demographic indicators across the various scenarios.

As regards population growth, all our scenarios for the period 2011–2030 take as their starting point historic data for the year 2010. In 2010 Mozambique's population was some 22.4 million in total, of which almost 31% lived in urban areas; average household size was 4.3, average annual population growth was 2.5%, with urban population growing 3.2% per year. Subsequently, all scenarios assume population growth to gradually decrease over time. In our Reference scenario we expect average annual population growth to decrease to 2.0% in 2030. In our Reference Low and High scenarios, we assume this number to be 0.4 percentage point higher and lower, respectively; the Extractive scenario is identical to the Reference High for all demographic indicators (see Table 3). As a result, by 2030 total population size is expected to be in the range of 33–37 million people, with 34.8 million people in the Reference Medium scenario (see Fig. 3). Growth of urban population is expected to increase to 3.5% per year in 2030 in the Medium scenario; in the Low and High scenarios we assume this percentage to be 50% lower and higher, respectively (see Table 3). As a result, by 2030 the

[3] The very high growth rate in 2001 followed the very low growth rate of 2000 that was caused by severe flooding. The recent slowdown of GDP growth can be primarily attributed to a significant decline of mining sector growth, reflecting lower international commodity prices and transportation constraints that prevented the scaling up in coal production and exports (IMF, 2016).

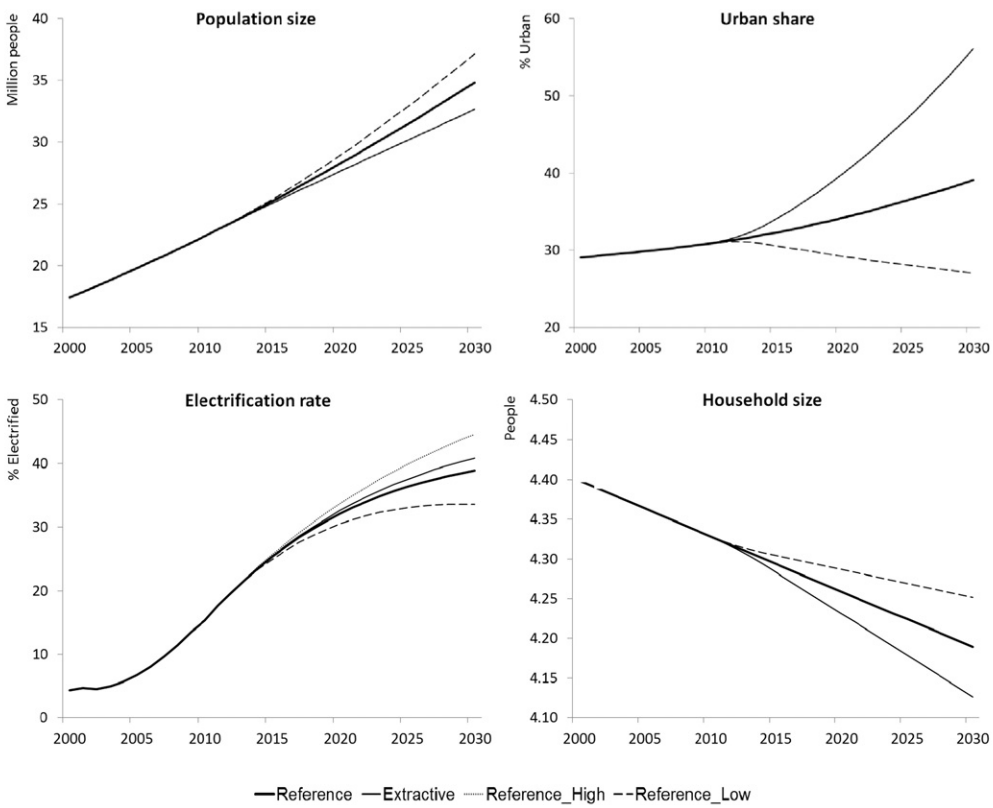

Fig. 3. Model results for population size, urbanization, electrification and household size.

percentage of urban population is expected to be in the range of 31–49%, with 39.1% urban people in the Medium scenario (see Fig. 3). This implies that in the Medium scenario the number of people living in cities in Mozambique by 2030 is as large as 60% of the entire population in 2010. Obviously this will not only reshape the urban landscape in Mozambique over the next 25 years, but also transform (residential) energy demand. Finally, in all scenarios we assume the average household size to gradually decrease under influence of income growth and urbanization, from 4.3 persons in 2010 to 4.13–4.25 persons in 2030 (see Table 3 and Fig. 3).

The extent to which the Mozambican population has access to electricity is expected to change rapidly as a result of intensive (rural) electrification programs and growing income levels. In our model the electrification rate is endogenously determined by combining information on electricity network expansion (number of new connections realized) with population growth dynamics as described above. In 2010 the national utility EdM realized 100.000 new connections. In our Reference scenario we expect this number to increase to 135.000 in 2015, and subsequently decrease to 100.000 in 2030. In our Reference Low and High scenarios, we assume that in 2030 respectively 70.000 and 130.000 new connections will be realized (see Table 3). Given population growth, this implies that in our model the (household) electrification rate is expected to increase from 15% in 2010 to 34% – 45% in 2030, with 39% in the Reference scenario (see Fig. 3). We assume transmission and distribution losses to remain at 5% as from 2011 (see Table 3).

3.3. Biomass model

To model future demand for fuelwood and charcoal we developed a simple biomass model, embedded within LEAP's overall accounting framework. Following micro-based evidence of household energy consumption patterns in developing countries (Barnes and Floor, 1999; Barnes et al., 2005), we adopt a nested model structure. First, we assume that total biomass consumption is merely determined by GDP per capita, thus considering substitution with modern energy forms (such as LPG and electricity) as function of relative prices a second-order effect (Leach, 1992). Second, we assume that the choice for one of the two dominant forms of biomass (fuelwood and charcoal) is driven by their relative prices as well as the urbanization rate (see Eq. (6)). More specifically, we first define the evolution of per capita biomass consumption B over time t according to a logarithmic S-shaped curve, as follows:

$$B_t = \alpha\left[1 + \beta e^{-\gamma Y_t^{Total}}\right], \tag{5}$$

where α is the initial value of B (in the year 2000), β is a constant (vertical shift of the curve), and γ the elasticity of B with respect to total GDP per capita, y_t^{Total}. The value for α is estimated on the basis of a combination of international data (IEA Energy Balances, 2010) and local household survey data (Atanassov et al., 2012; INE, 2009), and equals 10.5 GJ per capita. The values for β and γ are derived from a cross-country logarithmic panel data regression of biomass consumption on per capita GDP for the period 1971–2006, using IEA data for 74 countries with per capita values below US$3000; estimated coefficients for β and γ equal 0.0274 and 0.239, respectively. We refer to Table A.1 in the Appendix for details.[4] Next, we define the evolution of per capita consumption of charcoal C and fuelwood F as follows:

$$C_t = B_t\lambda_t \qquad \text{with} \quad \lambda_t = \lambda_{t-1}[(1+\rho)\gamma] \tag{6}$$

$$F_t = B_t[1-\lambda_t] \tag{7}$$

[4] The US$3000 cut-off criterion is chosen to avoid a potential bias in the estimated coefficients: the share of biomass in total energy consumption becomes in general very low in countries where GDP per capita exceeds US$3000; in our scenarios per capita GDP increases from about US$ 400 in 2010 to around US$1000 by 2030.

where λ is the share of charcoal in total biomass consumption, ρ is the inter-fuel substitution elasticity (i.e. between charcoal and fuelwood) and γ is the annual growth rate urbanization. Historical values for γ were derived from census data (INE, 2010), whereas values for λ in the initial year (2000) and ρ were derived from local household survey data (Atanassov et al., 2012; INE, 2009), with ρ set at 0.03. Future values for γ are taken from expected urbanization trends published by the UN in its World Urbanization Prospects (UN, 2008).

Finally, to allocate charcoal and fuelwood consumption across electrified and non-electrified households, we implemented the following assumptions. First, we assume that in 2000 all electrified households lived in urban areas, and that in 2011 the urban and rural electrification rates were, respectively 55% and 5% (IEA, 2013a). Second, we assume that 5% of total fuelwood consumption and 85% of total charcoal consumption is consumed by urban households with the remainder being consumed by rural households (Atanassov et al., 2012; Brouwer and Falcão, 2004; INE, 2009). Third, we assume that fuelwood and charcoal is consumed by, respectively, 33% and 80% of households in urban areas, while 10% of households in rural areas consume charcoal.[5] Finally, building on these assumptions we model future evolution of biomass consumption per electrified household b_t^{Elec} as a function of changes in the total biomass consumption (Eqs. (4)–(7)) as well as the change in urbanization rate U relative to the change in the electrification rate E, according to:

$$b_t^{Elec} = \Delta b_{t-1}\left[1+\frac{\Delta U}{\Delta E}\right]. \tag{8}$$

Biomass intensity per non-electrified household is subsequently derived from total biomass consumption not consumed by electrified households.

Fig. 4 illustrates the working of our biomass model within the LEAP framework for Mozambique. The left-hand side of Fig. 4 shows the evolution of per capita biomass consumption as function of per capita GDP, using actual values for Mozambique. Biomass currently is the principal energy source for the majority of Mozambicans, coming from an estimated 30.6 million hectares of forests and representing 80% of the energy consumed by households (Chambal, 2010). In our model, total biomass consumption declines with increasing GDP, following an inverted S-shaped patterns as defined by the logistic function of Eq. (4). As regards its composition, with rising levels of per capita GDP, consumption of charcoal increases at the expense of fuelwood consumption, under influence of rising income and urbanization – up to some income threshold level, after which it is substituted for modern energy forms such as LPG and electricity. The right-hand side of Fig. 4 demonstrates the substitution of fuelwood for charcoal across basic MOZLEAP model runs. In our baseline scenario ("Reference") the percentage share of charcoal in total biomass consumption in Mozambique increases from about 10% in 2000 (historical data) to almost 30% in 2030, thus decreasing the percentage share of fuelwood from about 90% to 70% over the same period. In the optimistic (high economic growth) scenario ("Reference_High") the expected percentage charcoal by 2030 is over 40%, in the pessimistic scenario ("Reference_Low") it still is expected to double from 10% in 2000 to 20% by 2030.

3.4. Fuel for road transport

We model fuel demand for road transport on the basis of the expected evolution of vehicle ownership over time, given the evolution of per capita GDP and population as described before. To this aim we developed again a simple logistic function that is embedded within LEAP's overall accounting framework. Following evidence from the top-down transport modeling literature in developing countries (Button et al., 1993, Medlock and Soligo, 2002) we assume the number of vehicles per 1000 people to be merely determined by GDP per capita, thus considering (relative) fuel prices a second-order effect. More specifically, we define the number of vehicles V per 1000 people at time t according to:

$$V_t = V_{t-1} * \left[1+\frac{\psi}{y_t^{Base}}\right]^{\Delta y_t^{Base}} \tag{9}$$

with parameter ψ denoting the elasticity of the change in vehicle ownership under influence of economic development. The value of ψ is derived from a cross-country logarithmic panel data regression of passenger car ownership on per capita GDP for the period 1971–2006, using data from the Worldbank Indicators database for 86 countries with per capita values below US$3000; the estimated coefficient for ψ equals 8.7. We refer to Table A.1 in the Appendix for details. In the absence of more detailed data we assume this parameter to apply equally to the evolution of passenger cars as well as trucks, motorcycles and tractors.

Next we calibrate our model by combining this approach with data on the annual evolution of registered vehicles in Mozambique and fuel consumption in the recent past, supplied by, respectively, the Mozambican National Institute of Road Transport (INATTER) and the Ministry of Energy (MoE, 2012). Fig. 5 summarizes the results of our methodology for estimating future demand for transport fuels across the various scenarios. It shows that the number of vehicles per thousand people is expected to grow from just over 10 in 2010 to about 29–38 in 2030, depending on the scenario. With increasing levels of per capita GDP, the implied GDP elasticity of vehicle ownership in our model is decreasing over time, from 2.5% in 2010 to about 1% in 2030. In short these numbers mean that the total number of vehicles in Mozambique is expected to increase four to five fold over the period 2010–2030, from just over 370,000 to about 1.6–2.0 million, depending on the scenario; of this total number of vehicles in 2030 in our model 63% consists of cars and 22% of trucks.

Mozambique has a large potential for the production of biofuel, given its climate and a vast amount of unused arable land. At this moment, biofuel production plays only a marginal role in the energy mix. However, the country has adopted a National Program for the Development of Biofuels to promote and use agro-energy resources for energy and food security. In doing so, the government also aims to encourage socioeconomic development and to reduce the country's dependence on fuel imports (Cuamba et al., 2013; Ecoenergy, 2008; IRENA, 2012). The program aims to progressively increase the proportion of biofuel in Mozambique's domestic liquid fuel mix in three phases. The pilot phase (2012–2015) is currently being implemented with a fuel blending mandate of 10% for bioethanol and 3% for biodiesel. An operational phase (2016 to 2020) will follow, with 15% bioethanol and 7.5% biodiesel blending and conclude with an expansion phase (2021-onwards) of 20% bioethanol and 10% biodiesel blending. In our scenarios, we include these phases, but take into account a 5-year delay to reflect the actual situation.

3.5. Demand scenarios from South Africa

South Africa's power utility (Eskom) has identified Mozambique as a potentially important supplier of electricity in its Integrated Resource Plan 2011 (SA Department of Energy, 2011) to help addressing its future supply-side challenges. Eskom is particularly interested in new hydropower from Mozambique, as the existing electricity generation mix in South Africa is carbon-intensive. Already, the Cahora Bassa (HCB) dam represents 40% of Eskom's carbon-free generation. One of the scenarios in the IRP is to use 2600 MW of power from Mozambique, including

[5] Note that rural electrification has a minor impact on switching of cooking fuel while the opposite is true for urbanization, which is a major driving force for the choice of cooking fuel.

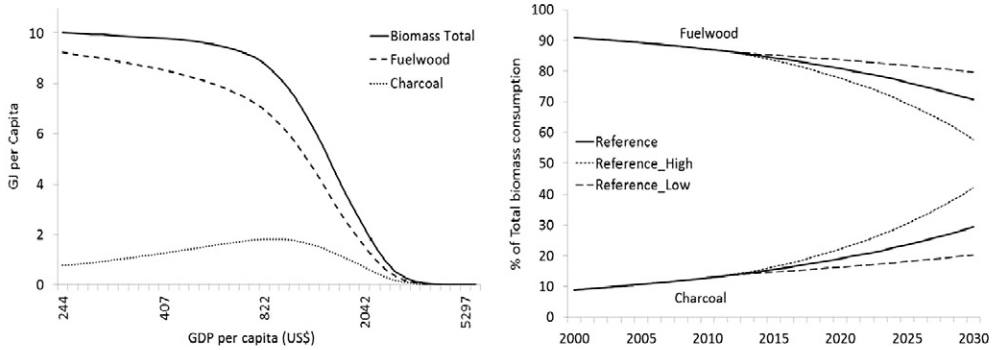

Fig. 4. Biomass consumption across scenarios – as function of GDP per capita (left) and its composition over time (right).

2135 MW from the new hydro projects. Electricity purchases from Natural Gas plants at the Mozambique-RSA border is not looked at in the 2011 Integrated Resource Plan (IRP). As of date, South Africa gets 92 MW from Gigawatt plant in Ressano-Garcia, and could get an additional 150 MW from Sasol's plant in the same area. According to the IRP, South Africa needs an additional 90 GW of generating capacity by 2030, mostly from renewables. Therefore, in our *Extractive Scenario,* we have modeled 3320 MW of capacity dedicated to Eskom, of which 1900 to 2100 MW would have to be firm.

3.6. Energy Intensity builder

As noted before, the LEAP accounting framework calculates (future) energy demand as the product of activity levels (such as GDP, population, physical production levels) and energy intensity per unit of activity. Therefore, the final building block of our model is an energy intensity builder that defines for each level of activity the corresponding energy intensity values over time. For the period 2000–2010 energy intensity values are calculated based on historical data regarding energy consumption and activity levels on a sector by sector basis. Subsequently, future energy intensity values for the period 2011–2030 are calculated on the basis of a variety of simple assumptions, again on a sector by sector basis. In this first integrated energy modeling and future planning study for Mozambique we deliberately apply simple and straightforward assumptions that do not vary in itself across scenarios. This choice is primarily motivated by our emphasis on exploring the potential impact of the expected unique surge in natural resources exploration in Mozambique on the country's energy supply and demand and economic growth prospects. Especially given Mozambique's current status as an extremely poor country with a rapidly expanding energy sector, this made us decide to leave a careful analysis of energy efficiency improvements in end-use sectors – although interesting and important – for future research.

A summary of our assumptions as regards future energy intensity trends across the various end-use sectors is presented in Table 4. We assume that in a poor country like Mozambique electricity consumption per household increases over time under influence of rising GDP, because growing household income leads to increasing demand for electric appliances such as refrigerators and air conditioning. As regards LPG, we assume that consumption per household increases over time under influence of rising GDP as well as the degree of urbanization, because growing household income leads to a shift towards modern cooking fuels, while in developing countries LPG is a typical urban fuel for logistic reasons. Furthermore, we assume that kerosene consumption per household decreases over time, because of a gradual 'autonomous' substitution towards more efficient and cleaner fuels like electricity and LPG. Finally, future charcoal and fuelwood intensities are derived from our biomass model. In short, we assume that total biomass consumption decreases under influence of increasing per capita GDP, while the share of charcoal in total biomass consumption increases with income and urbanization at the expense of fuelwood.

For the Agriculture sector we assume that energy intensity increases with about 20% over the course of 20 years, under influence of modernization and mechanization. In the Manufacturing sector we assume decreasing energy intensity growth, driven by the opposing forces of modernization and increasing energy efficiency. Under influence of increasing domestic natural gas production, we assume that the share of natural gas in this sector increases to 33% by 2030 at the expense of electricity and diesel shares; fuel oil plays a minor role. In the sector Commercial Services, we assume that electricity intensity increases with economic growth, while LPG consumption (in hotels and restaurants) again also positively depends on the degree of urbanization – following

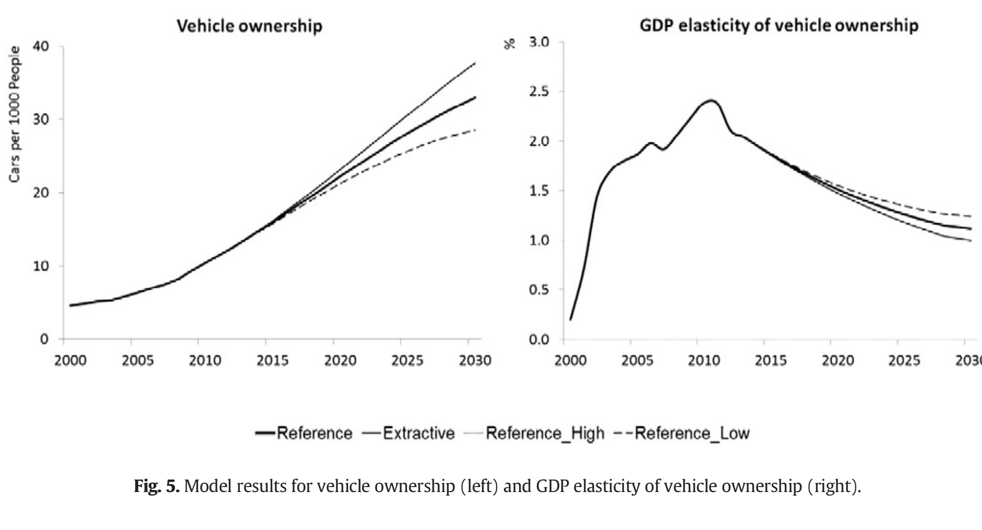

Fig. 5. Model results for vehicle ownership (left) and GDP elasticity of vehicle ownership (right).

Table 4
Key parameter values Energy Intensity Builder.

Sector	Fuel type	Characterization	Value or formula
Residential	Electricity	Increase	$I_t = I_{t-1}*(1 + (0,1*\Delta Y_t)$
	LPG	Increase	$I_t = I_{t-1}*(1 + \Delta Y_t) *(1 + \Delta U_t)$
	Kerosene	Decrease	-5% per year
	Charcoal	Increase	See Biomass model
	Fuelwood	Decrease	See Biomass model
Agriculture	Total	Increase	0.65 MJ/GDP (End year value 2030)
Manufacturing	Total	Increase	2.4%/year in 2011, gradually towards 0% / year in 2030.
Commercial Services	Electricity	Increase	1% per year
	LPG	Increase	$I_t = I_{t-1}*(1 + \Delta Y_t) *(1 + \Delta U_t)$
	Fuelwood, Charcoal	Decrease	-3% per year
Public Lighting	Electricity	Increase	$I_t = I_{t-1}*(1 + (0,5*\Delta Y_t)$
Government	Electricity	Increase	1% per year
Other sectors	Electricity	Increase	1% per year
MOZAL	Total	Constant	55.1 GJ/MT
Coal Mining	Electricity	Constant	27 kWh/MT
	Diesel	Constant	2 Liter/MT
Mineral Sands Mining	Electricity	Constant	600kWh/MT
	Diesel	Constant	2 Liter/MT
Tractors	Total	Increase	$I_t = I_{t-1}*(1 + (0.05*\Delta Y_t)$
Other Vehicles	Total	Decrease	$I_t = I_{t-1}*(1 + (-0.05*\Delta Y_t)$

the same logic as in the residential sector. Consequently, we assume a gradual substitution away from biomass consumption. Finally, we assume that electricity intensity for Public Lighting increases with economic growth. Also, in the sectors Government and Other we assume that electricity intensity will rise under influence of economic growth.

Energy intensity in the extractive industry is determined by constant values of electricity and diesel consumption per physical unit of production. We assume constant energy intensity values and fuel shares for the aluminum smelter MOZAL in the south of Mozambique, based on historical data, because we do not expect changes in its production process. Until recently MOZAL, which transforms imported bauxite into aluminum for export, was responsible for about three quarters of total electricity consumption in the country – and thus is a key feature of (future) energy planning and energy policy in Mozambique.[6] Actual values for the mining activities originate from a combination of indicative figures on open-cut coal mining and mineral sands explorations reported in the literature (Bleiwas, 2011; SEE, 2009) and from personal communications with local experts involved in mining activities in Mozambique. Finally, fuel efficiency in road transport is assumed to gradually increase over time under influence of economic development, which stimulates increasing import of newer and thus more fuel efficient vehicles. In contrast, we assume that fuel intensity for tractors increases because of the expected increasing use of heavy equipment as economic development proceeds. As regards the fuel mix, we assume a progressive use of biofuel in the domestic liquid fuel mix, adopting biofuel blending mandates from the government of Mozambique, taking into account a 5-year delay in accordance with the actual situation.

The energy intensities for the different economy sectors, measured as final energy consumption (in tons of oil equivalence(toe)) by the respective activity level (GDP in million US$), are presented in Table 5, for the Reference and Extractive scenarios. The Table shows that according to our model total energy intensity in Mozambique is expected to fall rapidly until 2030. This aggregate trend is driven by the combination of a few sector trends. First and most important, energy consumption per household is predicted to fall with 20–30% in the period 2010–2030. This decrease is mainly caused by the substitution of charcoal and modern energy types for the inefficient source of fuelwood. Given that in Mozambique the residential sector is responsible for the major part of energy consumption (see next section), this development obviously has a major impact on the evolution of aggregate energy intensity. Second, energy intensity levels are expected to fall relatively rapid in the Manufacturing and, especially, Extractive industry sectors. Manufacturing energy intensity changes mainly under influence of an increase in size and value added of manufacturing activities, starting from a very small base. The exceptionally high energy intensity level in the Extractive industry sector in 2010 is to be explained by its composition: the MOZAL aluminum smelter and the mineral sand exploration project in northern Mozambique (in Moma) – both of which operate highly energy-intensive processes. However, as the Table shows, energy consumption per unit of GDP in the Extractive industry sector is expected to fall rapidly over time because of the expected surge of natural gas and coal exploration, which broadens the production base. In contrast, because of our assumptions the other sectors (except Road Transport and Services) show a gradual increase of energy intensity over time.

4. Energy supply and demand projections

Using the modeling framework and assumptions as explained in the previous sections, we present in this section in more detail the future prospects of energy supply and demand in Mozambique, with its implications for the international energy markets.

4.1. Extractive industry developments

As was described in Section 3.1, we constructed extractive industry GDP on the basis of a bottom-up approach, based on information for individual projects in electricity production, gas exploration, coal mining and mineral sands mining. Below, we describe these projects in more detail, since they constitute a key element in our scenario paths regarding future energy supply in Mozambique.

As regards electricity generation, we consider in total 37 projects with a total capacity of almost 11.000 MW. Hydro is and remains to be the main source for electricity generation in Mozambique, with the greatest potential lying in the Zambezi River basin at sites such as Cahora Bassa North and Mphanda Nkuwa (Chambal, 2010). The existing capacity is around 2200 MW, of which 2075 MW is provided by the Cahora Bassa (HCB) dam. In total we consider in our model 15 hydro projects over the entire period 2000–2030, with a total capacity of about 7579 MW. In addition, in the period 2011–2030 we consider the construction of 12 natural gas fired power plants with a total capacity of 1114 MW as well as 6 coal fired power plants with a total capacity of 2150 MW. Finally, we include 101 MW from diesel generations and 1.2 MW solar power (cf. Cuamba et al., 2013). The latter reflects that

[6] For this reason, in our model we consider MOZAL to be part of the extractive industry sector, although it is strictly speaking not conducting extractive activities in Mozambique. Often, MOZAL is considered to be part of the Manufacturing sector.

Table 5
Energy Intensity indicators.

	Unit	2010	Reference				Extractive			
			2015	2020	2025	2030	2015	2020	2025	2030
Household	toe/hh	1.0	1.0	1.0	1.0	0.9	1.0	1.0	0.9	0.8
Agriculture	toe/10^6 US$	11.6	14.3	14.8	15.5	15.5	14.3	14.8	15.5	15.5
Manufacturing	toe/10^6 US$	748.9	546.6	403.0	318.0	266.5	544.9	464.4	354.3	282.1
Extractive Industries	toe/10^6 US$	3197.3	238.6	146.8	135.6	131.0	1996.2	601.4	325.8	270.2
Services	toe/10^6 US$	17.5	17.0	16.7	16.6	16.6	17.0	17.1	17.3	17.7
Government	toe/10^6 US$	8.4	8.9	9.3	9.8	10.3	8.9	9.3	9.8	10.3
Other Sectors	toe/10^6 US$	13.1	13.7	14.4	15.2	15.9	13.7	14.4	15.2	15.9
Road Transport	toe/10^3 vehicles	1.6	1.5	1.5	1.4	1.4	1.5	1.5	1.4	1.4
Total Economy	toe/10^6 US$	758.0	451.4	348.8	333.8	326.1	638.6	491.8	385.9	319.9

over the last decade PV solar energy has been gradually adopted in schools and health centers in rural areas, as well as in telecommunications businesses (Chambal, 2010). Next to its capacity, for each project we define its transformation efficiency, expected first year of production and merit order. As regards the latter, we divide the modeling period into four intervals and attribute merit order by expected first year of production in the basis of 5-years intervals, such that the value one represents existing power plants. Furthermore, the Reference scenario includes existing and very likely future power projects whereas the Extractive scenario includes all 37 power projects, including those whose realization is still fairly uncertain. We refer to Table A.2 in the Appendix for a detailed overview.

As regards the exploration of coal, our model includes in total 12 major coal mining projects with, in the Extractive scenario, a maximum total estimated annual production of 108 million tons by 2030. In the more moderate Reference scenario we consider 7 mining projects, which together account for 57 million tons per year by 2030. We refer to Table A.3 in the Appendix for details. For each mining project we define it's expected first year of production, the expansion of capacity over time, and the destination of its production (export versus electricity production).

As regards Natural Gas production, we of course start by modeling the existing natural gas exploration project by the South African company Sasol in the Pande & Temane gasfields. As noted before, the vast majority of natural gas produced from these fields is exported to South Africa through an 865 km pipeline. In addition, following the recent gas discoveries in the Rovuma Basin, we include the future construction of 12 so-called on-shore LNG trains plus 4 floating LNG units, each with a capacity of 5 million tons per year. Of this total capacity, we include 4 LNG trains (equivalent to 20 million tons per year) in the Reference scenario. Total gas production in the Reference scenario is then anticipated to reach just over 23 million tons per year as from 2018, which equals to about 1200 million GJ. In the Extractive scenario gas production could reach 65–80 million tons per year in the period 2025–2030, which equals to about 4200 million GJ per year in 2030. A detailed list is provided in Table A.4 in the Appendix .

Finally, on the demand side we model the evolution of several energy-intensive megaprojects, including five mineral sands mining projects and the MOZAL aluminum smelter. In the Reference scenario we assume that mineral sands mining only grows marginally from current levels to reach 1.2 million ton per year, with production confined to the existing Moma-Kenmare project. In the Extractive Scenario we include four new mining projects, with a total production level of 9.5 million ton by 2030. A detailed list is provided in Table A.5 in the Appendix . As regards the aluminum company MOZAL, we assume a constant physical production of 547 thousand tons per year in the Reference scenario, and an expansion to 728 thousand tons per year as of 2019 (often referred to as MOZAL-III) in the Extractive scenario. In doing so, we implicitly assumed that by 2019, the Center-South "Backbone" transmission line (CESUL) will be accomplished, such that the major power plants from the Zambezi Basin in the North of Mozambique are connected with the dominant economic center of the Maputo area in the South of Mozambique.

4.2. Energy supply and demand scenarios

Taken together, the aforementioned assumptions as regards extractive industry evolution in Mozambique, in combination with the previously described future development paths of GDP and population, drive energy production scenarios in our modeling framework. In the right-hand side of Fig. 6 we show that in the reference scenario we expect total primary production to increase from almost 14 million tons of oil equivalence (toe) in 2011 to 95 million toe in 2030. If Mozambique were to follow the extractive scenario development path, primary production could even increase to a level of over 180 million toe in 2030. This equals a 6 to 13-fold increase in primary energy production in less than 20 years. Clearly, this means that

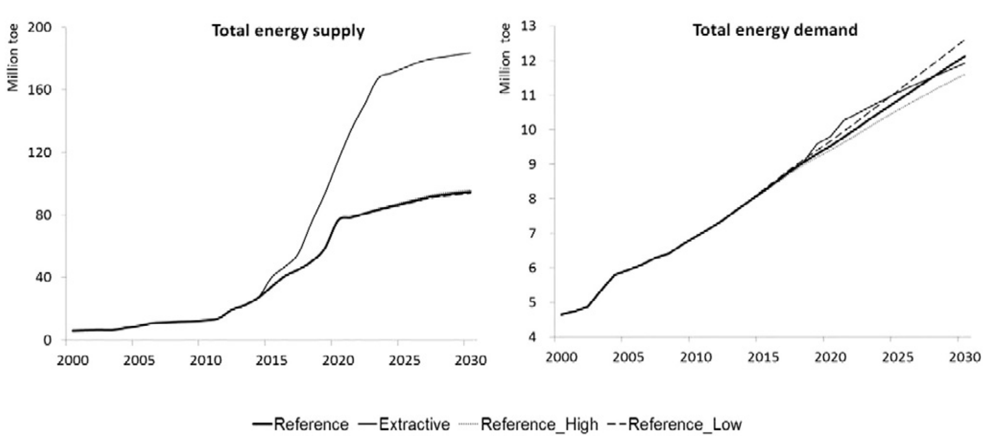

Fig. 6. Total energy supply (left) and demand (right) across scenarios.

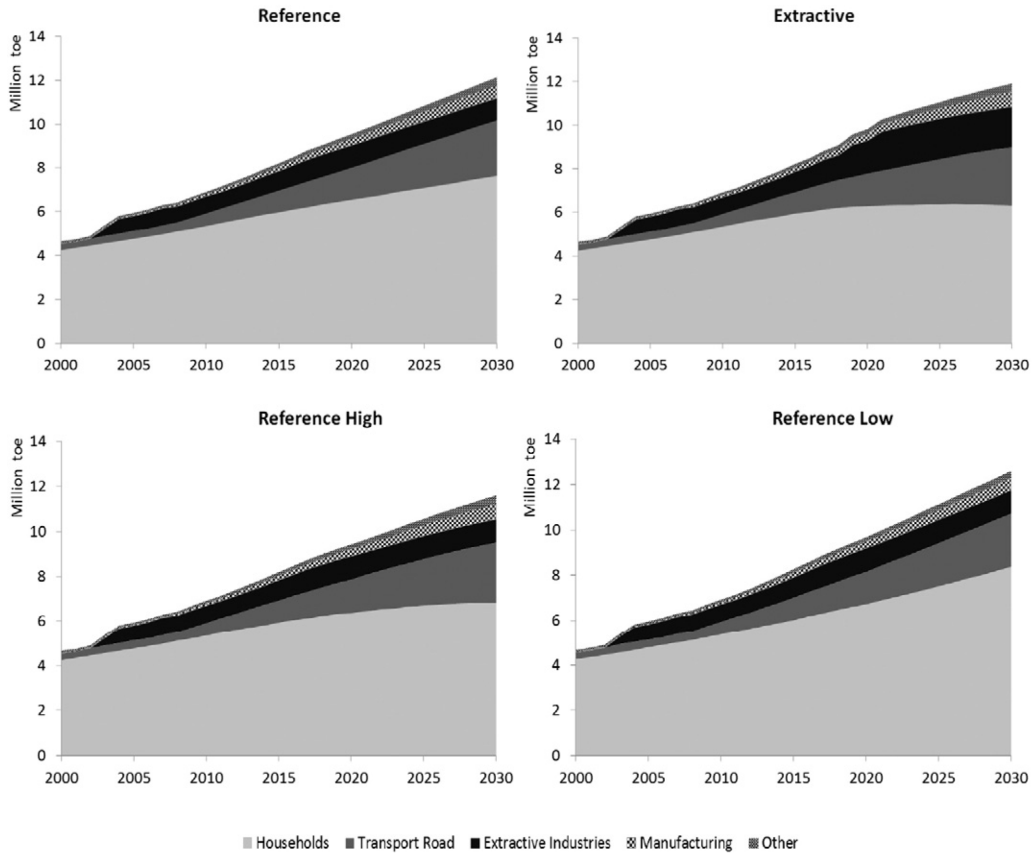

Fig. 7. Total final energy consumption by sector.

Mozambique will undergo no less than a revolution at the supply side of its energy sector.

The left-hand side of Fig. 6 pictures the evolution of aggregate final energy demand across the various scenarios. It shows that in the Reference scenario total energy demand is expected to increase to over 12 million toe in 2030. This is a 70% increase from the 2011 level of energy demand, and equivalent to an average annual increase of 2.8% as from 2011. If Mozambique were to follow the Reference Low development path, total final energy consumption is expected to reach almost 13 million toe in 2030, which equals an average annual increase of energy demand of 3.0% as from 2011. In contrast, the lowest level of energy consumption is to be expected if Mozambique were to follow the Extractive Scenario development path – with an estimated total final energy demand of 11.9 million toe in 2030, implying a 2.7% average annual increase over the period 2011–2030. The evolution of total final energy demand in the Reference High scenario is very similar to the Extractive scenario, notwithstanding differences in its composition.

It may appear at first sight somewhat counterintuitive that in the long run the Reference Low scenario yields a considerably higher level of aggregate energy demand than the Extractive or Reference High scenario – surely the latter scenarios include high economic growth and extractive industry expansion. Underlying data, however, clearly reveal that this result is to be explained entirely by the evolution of energy demand from the household sector – we illustrate this in Fig. 7. Given the relatively small size of the underdeveloped Mozambican economy, the residential sector is and remains responsible for a large part of total energy consumption in Mozambique (over 90% in 2000 and 50–60% in 2030). Fig. 7 shows that residential energy demand continues to grow relatively strong over time in the Reference Low scenario, whereas it decreases relatively rapid in the Extractive and Reference scenarios. Hence, the diverging energy demand patterns in the right-hand side of Fig. 6 are mainly caused by a straightforward scale effect: over time the number of households becomes much smaller in the Extractive and Reference High scenario than in the Reference Low scenario. This feature of our model of course follows from our assumption that population growth is inversely related to GDP growth (see Section 3.2). Consequently, it is in the high economic growth scenarios that the weight of the dominant household sector in driving total energy demand decreases most. In addition, there is an intensity effect: household energy intensity will decrease relatively rapidly over time in the high economic growth scenarios, given that our model assumes that households substitute away from inefficient biomass consumption towards the use of modern and more efficient energy types like LPG and electricity under influence of increasing per capita GDP (see Section 3).

As regards the non-residential sectors, Fig. 7 shows that road transport is the second largest sector in terms of energy demand across all scenarios. Also, it varies only to a relatively limited extent among the various scenarios, with an annual growth of about 7%. As a result, demand for fuel transport is expected to more than quadruple by 2030 as compared to 2010. Energy demand growth from the extractive industries is, of course, relatively strong in the Extractive scenario. According to this scenario, by 2030 energy demand of extractive industries equals over 1800 thousand toe, which is 80% more than in the Reference scenario. As noted before, to a large extent this is driven by the MOZAL aluminum smelter, which dominated the extractive industry sector until 2011. By 2030, MOZAL is still responsible for, about 50% of extractive energy demand in the Extractive Scenario; In the Reference scenario this is even 70%. in the Reference scenario. This is equivalent to 6–8% of total energy demand in the country in 2030, down from 11% in 2010.

Notwithstanding these high growth rates in demand, the true revolution in the Mozambican energy sector unfolds on the supply side, as noted before. To gain more insight in the consequences of the rapid emerging energy production levels in Mozambique, we provide

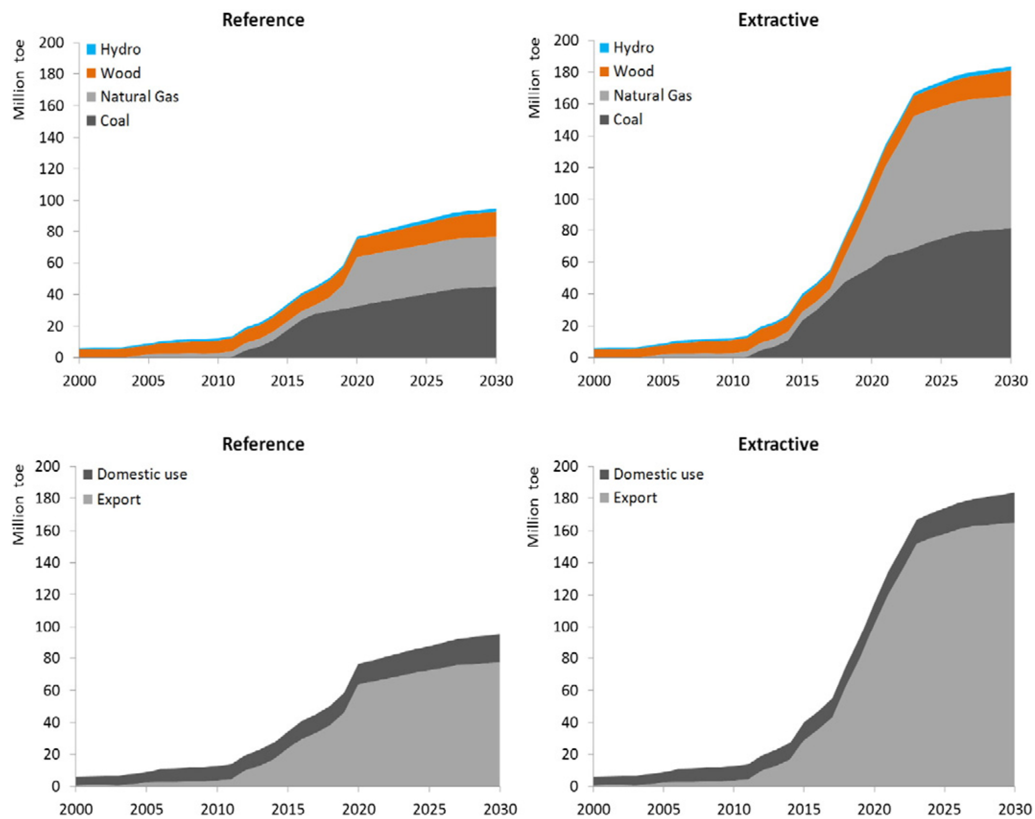

Fig. 8. Total primary energy production by source (top) and destination (bottom).

in Fig. 8 a breakdown of total primary energy production by source (top) and destination (bottom).

As can be seen from the upper part of Fig. 8, natural gas and coal together will make up for 80–90% of these production levels. In contrast, at the turn of the century 80–90% of total energy production in Mozambique consisted of wood, with hydro largely making up for the remaining 10–20%. Nevertheless, according to our scenarios total energy production from wood and hydro is expected to increase 60–90% over the next 15 years; yet the emergence of natural gas and coal are causing their relative shares in total energy production to decrease to about 3% (hydro) and 10% in 2030. The lower part of Fig. 8 shows that 80–90% of total energy production in Mozambique is expected to be destined for export. This is especially true for the natural gas and coal production, but to a lesser extent also for hydro, which continues to be exported to South Africa and other neighboring countries in the form of electricity (more on this below). The natural destination for coal is India, but Brazil is also expected to be a market for Mozambique's coal export (IEA, 2014). Of course, energy production from wood and solar almost exclusively serves the domestic market.

5. Conclusions

We developed the first comprehensive long-run scenario model of the emerging energy sector of Mozambique. Our analysis made use of the integrated modeling tool LEAP, to track energy consumption, production and resource extraction in all sectors of the Mozambican economy. Hence, we name the model MOZLEAP. It was our aim to introduce the model, and show its potential as a tool for energy planning and forecasting in the context of the emerging energy sector in Mozambique. We have described how the calibration of MOZLEAP is based on recently developed local energy statistics and international data for the recent past, as well as on information about the latest developments and future plans as regards the production and transformation of energy in Mozambique. We have shown how future GDP paths were built from a combination of macro trends and bottom-up developments in the extractive industry. Moreover, we presented the key mechanisms that drive our model results, including demographic and urbanization trends and cross-country based GDP elasticities with respect to biomass consumption, sector structure and vehicle ownership. We have developed four scenarios to evaluate the impact of the anticipated surge in natural resources exploration on energy supply and demand, the energy infrastructure and economic growth in Mozambique.

Our analysis suggests that until 2030, primary energy production is likely to increase at least six-fold, and probably much more. This is roughly 10 times the expected increase in energy demand; most of the increase in energy production is destined for export. As a result, Mozambique is rapidly developing into an important player at international energy markets; it may well become one of the leading global producers of natural gas and coal. We show that GDP growth rates in Mozambique are expected to accelerate and peak around 2020, which is largely driven by extractive industry expansion, most notably the expected start of large scale gas production in the Rovuma Basin, in the far north of the country. The International Monetary Fund recently predicted that as a result Mozambique's average growth rate during the first half of the next decade could reach over 20% per annum (IMF, 2016). Given the political, financial and technical uncertainties in developing these gas fields, we worked with more conservative assumptions, but nevertheless predict annual growth rates of over 10% in that period. Our projections imply that over the next decade the extractive sector will comprise about 10–20% of the total Mozambican economy, depending on the scenario.

Consequently, the rise of its energy sector in principle would allow Mozambique to transform the economy and accelerate its much needed development. The agriculture sector in Mozambique still is the source of livelihood for more than three-quarters of the population; agricultural productivity is so low that Mozambique remains a net food importer, and malnutrition is persistent. The country still ranks among the poorest countries in the world, with around half of the population living below the absolute poverty line and evidence of stagnation in the reduction of poverty rates over the last decade (Ross, 2014). Substantial tax revenues from extractive projects may help reduce the structural dependence of the Mozambican government on international aid to finance basic services and much needed investments in, among others, health, education and infrastructure.

But the recent past as well as our scenario results show that the opportunity to use natural resource wealth for developing an inclusive growth and development strategy in Mozambique poses major for its government. This is also demonstrated by the fact that the existing energy strategy of the Mozambican government has thus far been successful to a limited extend only. Inter alia, this strategy aims for promotion of new and renewable sources of energy, increased sustainable access to electricity and liquid fuels, diversification of the energy matrix, and institutional coordination and consultation with relevant stakeholders for better development (MoE, 2008). Our results show that the energy matrix indeed becomes much more diversified over time, but it also reveals that even with large scale development of the natural gas, coal and hydro potential, total energy production from traditional biomass will increase with at least 60% over the next 15 years. This is caused by the sustained dominance of the residential sector in total energy consumption, under influence of the relatively small size of the non-extractive economy in combination with high population growth rates. The latter also helps to explain our finding that even under optimistic assumptions the electrification rate is unlikely to exceed 45% by 2030. We have shown that PV solar energy has been gradually adopted, but also that it remains very small, percentage wise. A similar conclusion holds true for the production and consumption of biofuels. According to Chambal (2010), Mozambique has put in place a modern legislative framework for the energy sector in general and the power sector in particular. However, he also concludes that implementation and enforcement appear to lag behind considerably and some aspects are still unclear, particularly the contribution that the energy sector is to make to poverty reduction.

Clearly, the current expansionary strategy for the energy sector in Mozambique makes its economy increasingly vulnerable to a "resource curse", that may result from (a combination of) resource price volatility on international energy markets, Dutch disease effects, crowding out of productive investments, or discouragement of institutional and capacity building in public administration (Van der Ploeg, 2011). Over the last two years GDP growth already slowed down because of a decline in mining sector growth, which was caused by lower international commodity prices and transportation constraints – the delayed development of a new railway line and deep-water port that prevented the scaling up in coal production and exports. Large scale development of the country's hydroelectricity potential in the Zambezi river basin has now been delayed for years, due to uncertainty about necessary long-term demand agreements with neighboring countries in combination with interregional network constraints. The Mozambican government faces the challenge to balance the need for these and other natural resource related infrastructure with the need for investments in health, education, water and agricultural productivity. Moreover, there is a need to weigh the planned expansion of thermal electricity generation capacity on the basis of coal and natural gas, against development of the country's large hydroelectricity potential, also in view of the environmental consequences of both strategies.

Until recently, megaprojects in the extractive sector contributed little to fiscal revenues, but according to the IMF, by 2020 resource revenues could exceed 40–50% of total revenues and would last for over 35 years (Ross, 2014). This would require major improvement of public financial management to ensure that resource wealth is used efficiently and transparently, including the adoption of new fiscal rules that can help safeguard the annual budget from price volatility (Van der Ploeg, 2014). In addition, it becomes increasingly clear that there is an enhanced need to manage existing regional imbalances in resource wealth, which is concentrated in the far north of the country whereas other economic development and political power is concentrated in the far south of the country. Since recently, Mozambique is facing increasing local institutional and political instability, which has its roots in the country's devastating civil war during the period 1977–1992. This includes returning instability in the central region of the country led to attacks by armed insurgents on newly developed mining and transport infrastructures as well as on main roads that connect the northern and southern regions of the country. Large scale natural resource exploration is likely to increase these tensions, which underlines the importance of developing an inclusive growth strategy that covers all regions of the country. Certainly, this is much more easily said than done, because institutional capacity in Mozambique is still weak in many areas. To address the formidable list of challenges described above will therefore require continued improvement of the quality of public administration. Hence, the key question, therefore, seems to be whether the expected natural resource boom will encourage or frustrate the required improvements of local institutional capacity.

Acknowledgements

The first author wants to thank the Swedish International Development Agency (SIDA) for financial support, provided as a research grant under the Energy Science and Technology Research Program (EnerPro) of the Eduardo Mondlane University in Maputo, Mozambique. Part of the data used in this paper were collected as part of the EuropeAid/127640/SER/MZ project "Capacity Building in Energy Planning and Management in Mozambique", in which both authors participated, together with David Nadaud. We would like to thank David Nadaud for his excellent help in collecting local energy statistics, as well as his valuable contributions to the work and discussions that preceded this paper. Also, we would like to thank staff from the then Mozambican Ministry of Energy, Directorate of Studies and Planning, as well as Nazario Meguigy and Marco Coutinho for their valuable help in data collection. Finally, we would like to thank Henri L.F. de Groot as well as Ronald Huisman and other participants of the 2014 workshop of the Benelux Association of Energy Economics (BAEE) for constructive comments. The content of the article does not necessarily reflect the views of any institution that was involved in the construction of the energy statistics that we used for this paper. All remaining errors are our responsibility alone.

Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.eneco.2016.08.010.

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